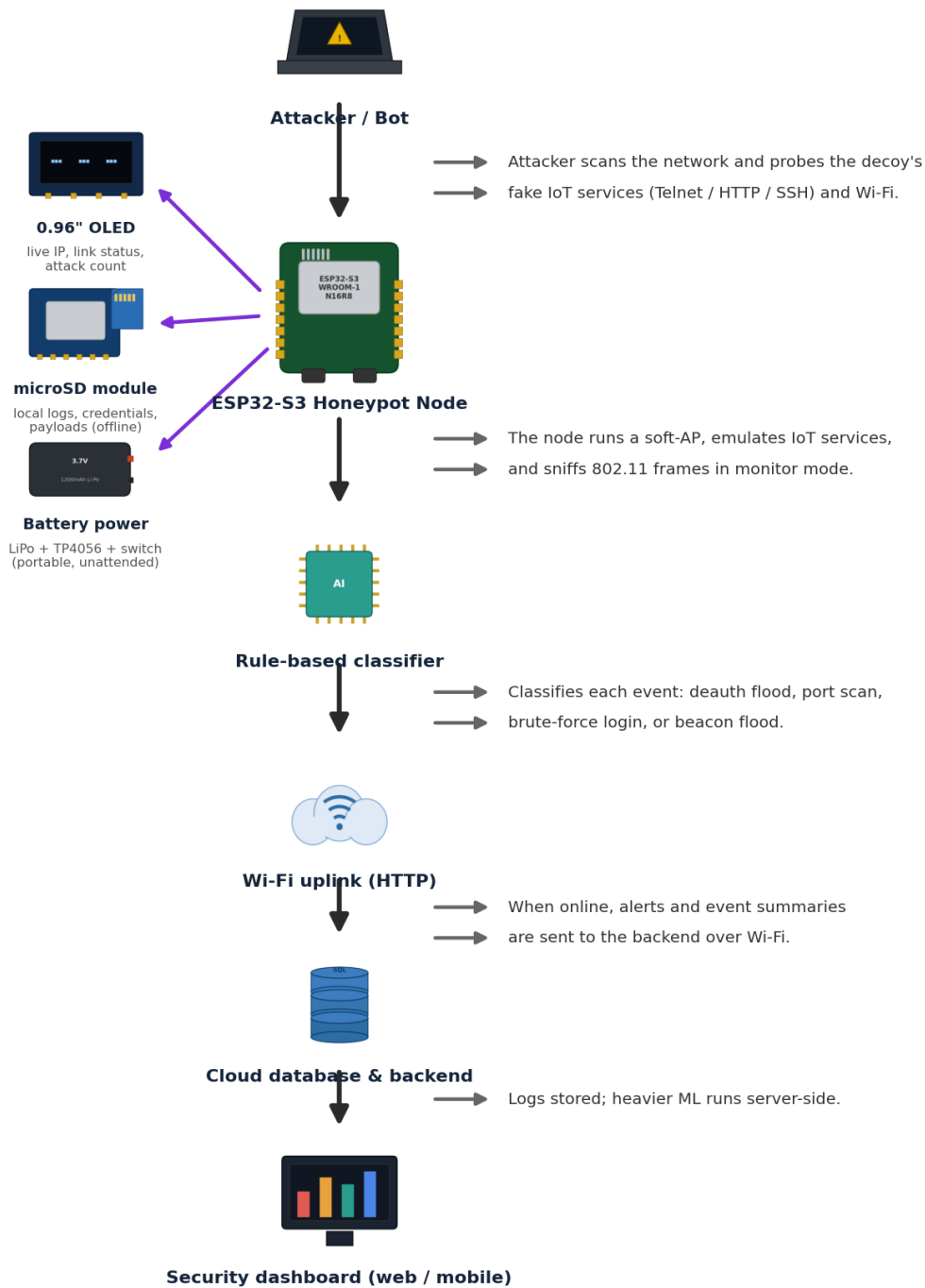


Project Block Diagram

System Architecture of the ESP32-S3 Network Honeypot Node

System Architecture of ESP32-S3 Network Honeypot Node



Device / Product Comparison Table

Feature	Proposed: ESP32-S3 Honeypot Node	Thinkst Canary (commercial)	Raspberry Pi Honeypot (T-Pot / Cowrie)
Approximate Cost (USD)	~\$25	\$1,000+ / yr (subscription)	~\$70
Power Consumption	Low (battery)	Moderate (mains)	High (mains)
Battery Powered / Portable	Yes	No	No
Fake IoT Service Emulation (Telnet/HTTP/SSH)	Yes	Yes	Yes
Wi-Fi Deauth / 802.11 Frame Capture	Yes	No	Limited
Port-Scan Detection	Yes	Yes	Yes
Brute-Force Credential Capture	Yes	Yes	Yes
On-Device Attack Classification	Yes	Yes	Limited
Local Offline Logging (microSD)	Yes	No	Yes
Real-Time On-Device (OLED) Status	Yes	No	No
Cloud Dashboard / Alerts	Yes	Yes	Yes
Managed Console / Vendor Support	No	Yes	No
High-Interaction Realism	Limited	Yes	Yes
Open-Source / Customizable	Yes	No	Yes
Ease of Deployment	Easy	Easy	Moderate
Suitable for Home / Small Networks	Yes	Limited	Yes
Suitable for Enterprise	Limited	Yes	Limited

The proposed node trades the high-interaction realism and vendor support of a commercial appliance for very low cost, battery portability, low power, and link-layer (Wi-Fi) attack capture, making it well suited to home and small-network deployment.

Available Resources

Hardware and suppliers

- ESP32-S3-DevKitC-1 (N16R8, 16 MB flash / 8 MB PSRAM) — main controller.
- 0.96" I²C OLED (SSD1306), Micro SD card module + 16/32 GB card.
- TP4056 Type-C charger, 3.7 V 1200 mAh Li-Po cell, SPDT slide switch, MB102 breadboard, jumper wires.
- Local suppliers: RoboticsBD, Robotics Shop BD, DreamRC, RoboDoc BD (see project component list).

Software and tools

- Firmware: Arduino IDE / ESP-IDF (C / C++), ESP32 Wi-Fi promiscuous (monitor) mode API.
- Attack generation for testing: **nmap** (port scans), **aircrack-ng/aireplay-ng** (deauth, beacon flood), **hydra** (brute force), Kali Linux on a laptop / Raspberry Pi.
- Logging and dashboard: microSD (FAT), and a cloud back end / dashboard (e.g. ThingsBoard or Blynk) for alerts.

Datasets and references

- IoT-DH honeypot DDoS dataset — <https://doi.org/10.1016/j.dib.2024.110496>
 - Survey of honeypots/honeynets for IoT — <https://arxiv.org/abs/2108.02287>
 - ESP32 Wi-Fi deauthentication detection reference — <https://doi.org/10.3390/fi16060200>
 - Thinkst Canary — <https://canary.tools> T-Pot honeypot platform — <https://github.com/telekom-security/tpotce>
-
- Project files (Google Drive) — <https://drive.google.com/file/d/1CNUPtEoeu1RoBv5bDcmdIiKfyPBw8zYA/view?usp=sharing>